SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10 **Total Marks:20**

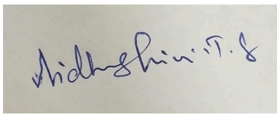
Annexure A

CLASS TEST

Criteria	Understanding of Task *(2)*	Procedure & Execution *(2.5)*	Observation & Result *(2)*	Viva Voce / Conceptual Response *(2)*	Time Management & Presentation *(1.5)*	***Total(10)***
Weightage	*20%*	25%	20%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						
Q9						
Q10						

Code of Conduct:

I Vidhushini.T.S (192511061) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS: **Total Marks: 10x2 =20**

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formals[L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set x = {4,5,6,7,9}, y = {8,3,7,5,6}

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, l5]

Q1. Define Data Mining and list any two basic data mining tasks. Explain the ~~sp~~steps involved in KDD process and justify the role of data preprocessing in it.

Data Mining: Process of finding useful patterns and knowledge from large datasets.

Two tasks: Classification, clustering

KDD Steps:

1. Data Selection
2. Data preprocessing
3. Data Transformation
4. Data Mining
5. Pattern Evaluation
6. Knowledge Presentation.

Q2: What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values.

Data Cleaning: Process of correcting or removing incorrect, incomplete and noisy data.

Two quality issues: Missing Values, Noisy data.

Missing values: Replace with mean, median, mode.

Noisy data: Handle using Binning.

Q3: Define Data integration and list two challenges associated with it.

Data Integration: Combining Data from multiple sources into one consistent dataset.

Challenges:

↳ Schema Differences

↳ Data redundancy.

Q4. What is covariancy analysis? List 2 covariance formals.

<u>Covariance</u> : Measures direction of relationship between two variables.

Positive covariance and negative covariance.

<u>Population</u> : $\boxed{Cov(x, y) = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{N}}$

<u>Sample</u> : $\boxed{Cov(x, y) = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{n-1}}$

Q5. Suppose two stocks A and B have following values in one week (2,5) (3,8), (5,10) (4,11), (6,14).

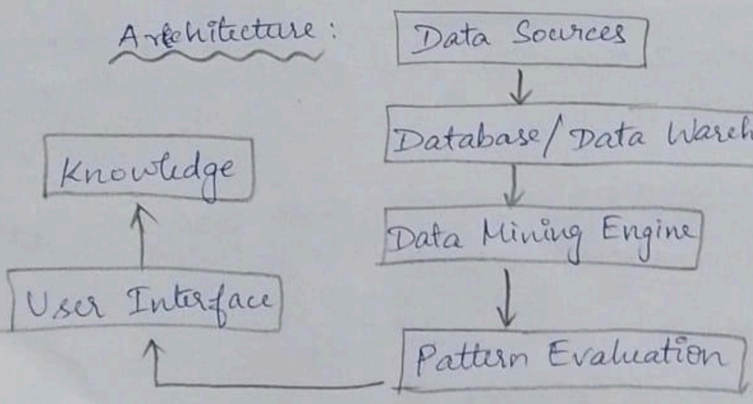
<u>Covariance</u> = $Cov(A, B) = E(A \cdot B) - E(A)E(B)$

$A = 2, 3, 5, 4, 6 \quad B = 5, 8, 10, 11, 14$

$\boxed{\text{Covariance} = 4.}$

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4].

<u>Architecture</u> :



Components:

↳ Data Sources: Provide raw data

↳ Database: Stores data

↳ Mining Engine: Finds Pattern

↳ Pattern Evaluation: Indentifies useful patterns

↳ User Interface: Display results.

Q7. Find the covariance for following data set

$X = \{4, 5, 6, 7, 9\}$, $Y = \{8, 3, 7, 5, 6\}$

$X = 4, 5, 6, 7, 9$

$Y = 8, 3, 7, 56$

Means: $\bar{X} = 6.2$ $\bar{Y} = 5.8$

Population Covariance $= -0.36$.

Covariance $= -0.36$, so the relationship is negative.

Q8. Describe the major steps in data preprocessing. Given a dataset with missing values, outliers, inconsistent formats, propose a systematic preprocessing pipeline and justify the sequence of steps.

<u>Major Steps</u>:

1. Data Cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction
5. Data Discretization.

Pipeline:

Raw Data → Missing Values → Outliers → Inconsistent Formats → Transformation → Reduction → Preprocessed Data.

Q9. Hospital is analyzing patient age data to identify broad health trends. Sample DataSets: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4].

Total = 27 ages. For 3 bins: 27/3 = 9.

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Using bin mean: Bin 1 → 18

Bin 2 → 28.67

Bin 3 → 43.78

Q10. Explain data discretization and its importance in data mining. Compare equal-width & equal-frequency binning with examples.

Data Discretization: Converting continuous numerical data into intervals/categories.

Equal width: Divide range into equal intervals

Example:

0-25, 25-50,

50-75, 75-100

Equal-Frequency: Divides data into bins containing approx. equal no. of values.

Equal width	Equal-Frequency
Same interval width	Same number of values
Range-based	Data-count based
Can be affected by outliers	Better for uneven data